

BEGIN

Let vehicle-type, user-cat, has-permit, is-emergency: CHAR

capA, capB, capC: INT

Set capA = 20

capB = 40

capC = 15

PRINT "Enter Vehicle type (C: Car, B: Bike, V: Van): "

INPUT vehicle-type

PRINT "Enter User Category (F: Faculty, S: Student, G: Visitor): "

INPUT user-cat

PRINT "Has Valid Permit? (Y/N): "

INPUT has-permit

if (vehicle-type != 'C' AND vehicle-type != 'B' AND vehicle-type != 'V') OR (user-cat != 'F' AND user-cat != 'S' AND user-cat != 'G') OR (has-permit != 'Y' AND has-permit != 'N') then

PRINT "Error: Invalid Input "

else

if (has-permit == 'N') then

PRINT "Is Emergency vehicle? (Y/N): "

INPUT is-emergency

else

is-emergency = 'N'

end if

```
if (has_permit == 'N' AND is_emergency == 'N') then  
    PRINT "Access Denied"
```

```
else
```

```
    if (user_cat == 'F') then
```

```
        if (capA >= 1) then
```

```
            capA = capA - 1
```

```
            PRINT "Access Approved: Assigned to Zone A"
```

```
        else
```

```
            PRINT "Access Denied: Zone A is full"
```

```
        endif
```

```
else
```

```
elseif (user_cat == 'S') then
```

```
    if (vehicle_type == 'V') then
```

```
        if (capC >= 2) then
```

```
            capC = capC - 2
```

```
            PRINT "Access Approved: Redirected to Zone C"
```

```
        else
```

```
            PRINT "Access Denied: Zone C has insufficient  
                space"
```

```
        endif
```

```
    else
```

```
        if (capB >= 1) then
```

```
            capB = capB - 1
```

```
            PRINT "Access Approved: Assigned to Zone B"
```

```
        else
```

```
            PRINT "Access Denied: Zone B is Full"
```

```
        endif
```

```
    endif
```

```
else if (user_cat == 'G') then
    if (vehicle_type == 'V') then
        if (capC >= 2) then
            capC = capC - 2
            PRINT "Access Approved: Assigned to Zone C"
        else
            PRINT "Access Denied: Zone C has insufficient
                space"
        endif
    else
        if (capC >= 1) then
            capC = capC - 1
            PRINT "Access Approved: Assigned to Zone C"
        else
            PRINT "Access Denied: Zone C is Full"
        endif
    endif
endif
endif
endif
endif
END
```